

Technical Progress Report

Project name:	Biped robot		
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1. Executive Summary

This report summarizes the technical progress of the biped robot project this week. The team completed the optimization of the power supply system: replaced the unstable DuPont wires with a dedicated DC cable to eliminate voltage/current fluctuations, and adopted a new high-capacity battery to ensure stable power output for servos. With the upgraded power system, all joints of the robot have achieved normal and reliable movement. The next step is to conduct walking tests next week.

2.Objectives & Scope

- Replace unstable DuPont power wires with a dedicated DC cable to ensure stable voltage and current for servos.
- Integrate a new high-capacity battery to provide continuous, reliable power supply.
- Verify the stability of the upgraded power system during servo operation.
- Achieve normal movement of all robot joints.
- Prepare for walking tests in the next stage.

3. Technical Work Completed

3.1 Power Supply System Optimization

Replaced the original unstable DuPont connection with a new dedicated DC power cable for servos, eliminating voltage/current fluctuations during movement.
Purchased and integrated a new high-capacity battery to provide sufficient, steady power for long-time operation.
Completed wiring layout and voltage matching for the whole robot system.

3.2 Joint Movement Debugging & Verification

Successfully debugged and verified normal movement of all robot joints (hip, knee, ankle, etc.) with the upgraded power system.
Confirmed no servo jitter, voltage drop or automatic reset during continuous joint operation.
Completed basic motion sequence testing for each joint to ensure responsive and accurate control.

3.3 Upper Body Structure Update

Completed the re-modeling of the robot's upper body, designed the mounting positions for the ESP32 board, camera and battery.
The new structural parts are under 3D printing and will be assembled this week.

4. Observations & Findings

- The original DuPont wires caused unstable voltage and current, leading to servo jitter and unreliable movement.
- The new DC cable significantly improves power stability and reduces signal interference.
- The new high-capacity battery maintains steady voltage under load, supporting continuous joint operation.
- All joints now move normally and responsively with the optimized power system, laying a solid foundation for walking tests.
- The re-designed upper body structure provides reasonable space for hardware integration, improving the robot's overall stability.

5. Challenges & Roadblocks

- Unstable power supply from DuPont wires directly affected servo performance and joint movement.
- Insufficient capacity of the old battery led to quick voltage drop and abnormal servo reset.
- The new upper body structure requires precise assembly to ensure hardware fixation and wiring safety.

6. Next Steps & Recommendations

- Assemble the newly printed upper body structure, fix the ESP32 board, camera and battery firmly.
- Conduct comprehensive joint movement tests to confirm long-term stability.
- Attempt walking tests next week to verify the robot's balance and motion control.
- Optimize the walking gait algorithm based on test results.
- Continue troubleshooting any remaining servo control issues.

7. Conclusion

This week, the team focused on power system optimization and joint movement debugging. By replacing DuPont wires with a DC cable and adopting a high-capacity battery, we solved the power instability problem, and successfully achieved normal movement of all robot joints. The re-designed upper body structure is under printing and will be assembled soon. The project has reached a key milestone, and the next stage will focus on walking tests to realize the robot's basic walking function.